APPENDIX O

SECONDARY DATA USE FORM

Complete this form if research involves use of existing data or data being collected for non-research purposes.

Use this form if your study is limited to analysis of existing data, documents, records, or specimens.

NOTE:

- Please make sure that the information provided on this form is consistent with the information included on the IRB Application, which you must also complete as part of your IRBNet submission.

1. RESEARCH PURPOSE

1.1. Briefly explain the purpose of research, the research questions, and the potential value.
This study (IRB Protocol #2446127) investigates whether the categorical and hierarchical structure of Library of Congress Subject Headings (LCSH) — the primary controlled vocabulary used in library subject cataloging — corresponds to measurable patterns in human semantic cognition. The central research question is: do the sense boundaries encoded in LCSH function as psychologically meaningful partitions of semantic space? The study introduces the concept of "cognitive warrant" as a complement to the established concept of "literary warrant," which governs when new subject headings are created. Cognitive warrant holds that a categorical boundary in a knowledge organization system has validity to the extent that it corresponds to a measurable discontinuity in human judgments of semantic similarity or processing difficulty. The potential value of this research is twofold. If LCSH boundaries are confirmed to have cognitive warrant, this provides empirical support for their role in subject access design. If they are not, the framework identifies specific headings or heading types where literary warrant is strong but cognitive warrant is weak— a finding that directly informs cataloging rule design and the evaluation of knowledge organization systems. Either outcome contributes a validated, replicable methodology for empirically evaluating controlled vocabulary structures using tools from psycholinguistics and computational linguistics.

1.2. What are the study aim(s) question(s) or hypotheses this activity is designed to answer.
The pilot study addresses three research questions. The secondary data analysis — using Trott and Bergen's publicly available dataset (Trott, S., & Bergen, B. (2023). Word meaning is both categorical and continuous. Psychological Review, 130(5), 1239–1261. https://doi.org/10.1037/rev0000420) merged with LCSH structural variables derived by the principal investigator — provides proof-of-concept evidence bearing primarily on Pilot RQ2 and Pilot RQ3. Pilot RQ1 is addressed by the librarian consultation component described in the main application. Pilot RQ2 (Structural Correspondence): Do BERT-derived semantic distances between word sense pairs and LCSH-derived hierarchical distances between their corresponding headings exhibit the expected directional relationship — that is, do sense pairs that are semantically closer also tend to have hierarchically

closer headings in the LCSH graph? Pilot RQ3 (Behavioral Feasibility): Do pilot participants' response time patterns (Survey 1) and heading appropriateness ratings (Survey 2) show directional trends consistent with the cognitive warrant framework, justifying the design of a fully powered follow-on study? The secondary data analysis tests the following three pilot hypotheses at the sense-pair level using Trott and Bergen's existing response time data. All hypotheses are evaluated descriptively; inferential testing is deferred to the fully powered follow-on study. Pilot H1 (Continuous Similarity Effect): Greater BERT-derived semantic distance between paired stimuli will be positively associated with response time (log-transformed; Survey 1) and negatively associated with heading discrimination scores (Δr; Survey 2), reflecting a continuous influence of distributional semantic similarity on implicit processing difficulty and explicit evaluative judgment. Pilot H2 (Categorical Boundary Effect): Controlling for continuous BERT-derived semantic distance, stimulus pairs mapped to different LCSH subject headings (same_heading = 0) will exhibit longer response times and lower discrimination scores than pairs mapped to the same heading (same_heading = 1). Pilot H3 (Structural Correspondence): Among word sense pairs whose LCSH headings share a common broader-term ancestor within three hierarchical levels (same_nbhd = 1), BERT-derived semantic distance and LCSH hierarchical distance (shortest-path length via lowest common ancestor) will be positively correlated — that is, pairs with greater distributional semantic distance will tend to be mapped to headings that are farther apart in the LCSH graph.

2. RESEARCH PROCEDURES INVOLVED IN THE SECONDARY DATA ANALYSIS

2.1. Provide a complete description of your study design and all the study procedures that you will perform. For example:

- How will data be obtained?
- Are the data identifiable?
- Will any member of the study team have access to the code that links identifiers to subjects?
- Did subjects provide consent for original data?
- If a HIPAA waiver is needed to access existing private data, make this clear.

Study design: The secondary data analysis applies a cross-classified hierarchical linear model to an existing, published dataset. The analysis is purely computational; no new data are collected from human participants, and no interaction with participants occurs. How data will be obtained: The dataset was collected by Trott and Bergen (2023) and published in full on GitHub at https://github.com/seantrott/trott_ph_amb under an open license. The data are freely and publicly available without registration, credentialing, or data use agreement and will be accessed by the principal investigator directly from this repository. Identifiability: The published dataset contains no direct personal identifiers (no names, addresses, contact information, or government ID numbers). Participants are represented by anonymous numeric subject codes assigned during the original study. The dataset contains limited indirect identifiers: participant age, self-reported gender, and native language status. These variables are included in the original dataset as covariates; the principal investigator will use them only as statistical controls and will not use them to attempt to identify any individual. Access to linking code: No code linking subject identifiers to named individuals exists in the published dataset or is accessible to the principal investigator. The anonymous subject codes in the published data cannot be reverselinked to individuals. Consent in the original study: Participants in Trott and Bergen's (2023) study provided informed consent under that study's IRB protocol. By publishing their data under an open license on OSF and GitHub, Trott and Bergen have made the data available for secondary research use. The dataset's OSF repository explicitly designates the data as openly available for reuse. HIPAA: Not applicable. The data contain no protected health information. Analytical procedures: The principal investigator will merge the published response time data with LCSH structural variables (same_heading, hier_distance, same_nbhd) derived from a separate computational pipeline using the Library of Congress Linked Data Service. The merged dataset will be analyzed using cross-classified hierarchical linear models in R (lme4 package), with log-transformed response time as the dependent variable and LCSH structural variables as fixed-effect predictors alongside BERT cosine distance. No data from the published dataset will be modified, imputed, or collected anew.

3. EXISTING DATASET

3.1. Are the data existing at the current time?

☒ Yes
☐ No

Who/what is the source from which the data set will be obtained?
(Data should not be obtained prior to IRB approval or exemption)

The dataset was collected by S. Trott and B. Bergen as part of their 2023 study, Word meaning is both categorical and continuous. Psychological Review, 130(5), 1239–1261. https://doi.org/10.1037/rev0000420.

3.2. Describe the process for gaining permission to use the data set, including any requirements, agreements, or credentials necessary to access the data. Attach a letter of cooperation or agreement for the data access.
The dataset was collected by Trott and Bergen (2023) and published in full on GitHub at https://github.com/seantrott/trott_ph_amb under an open license. The data are freely and publicly available without registration, credentialing, or data use agreement.

3.3. Does the original file contain direct or indirect personal identifiers?

☒ Yes
☐ No

Direct personal identifiers include information such as: name, address, telephone number, social security number, identification number, medical record number, license number, photographs, biometric information, etc.

☐ Yes
☒ No

Indirect personal identifiers include information such as: race, gender, age, zip code, IP address, major, etc.

☒ Yes
☐ No

3.4. If you answered 'YES' to either in 3.3., please respond to the following:

1. Please list the personal identifiers (direct and indirect) that will be included in the data set:

The dataset includes the following indirect identifiers: a 10-digit alphanumeric randomized subject ID (randomly assigned and not reverse-linkable to any named individual; included solely to support within-subjects statistical analysis), self-reported gender, native English speaker status (yes/no), mobile device used (yes/no), age, and their response to two bot check questions.

2. Will you remove the identifiers from the data set or otherwise maintain and analyze the data in such a manner that individuals cannot be identified either directly or indirectly through identifiers linked to participants? (A de-identified data set refers to original data that has been stripped of all elements that might enable a reasonably informed and determined person to deduce the identity of the participant.)

☒ Yes
☐ No

If answered 'NO', please provide brief justification:
Click here to enter text.

3.5. Please clarify the following regarding the consent process (Choose 1 below):

☒ Consent for use of the data for this purpose was obtained at the time of the data were originally collected. Attach the original consent document.

☐ Consent will be obtained from each participant group.
If you plan to obtain consent from the participants, you must complete and attach either the informed consent form (using the DU IRB Template online) or the exempt information sheet (if research is exempt).

☐ Waiver of consent is requested.
Please complete the Appendix A. Be sure to select the option for a "Waiver or Alteration of Consent" and to provide sufficient justification in each of the boxes provided.

Please describe briefly how consent will be obtained, or if applicable, why consent will not be obtained. Participants in Trott and Bergen's (2023) study provided informed consent under that study's IRB protocol. The original consent document from the Trott & Bergen study is not separately available; the authors' open licensing of the dataset on GitHub (detailed in section 2.1) constitutes their permission for secondary use, an authorization that extends to the current study. Additionally, and independently, secondary research on publicly available data is exempt under 45 CFR 46.104(d)(4)(i) regardless of the consent language in the original study, because the regulatory exemption does not condition secondary use on participant authorization for such use.

3.6. Explain how researchers will maintain confidentiality of the data.
Describe how researchers will protect data against disclosure to the public or to other researchers or non-researchers. Other than members of the research team, explain who will have access to the data (e.g., sponsors, advisors, government agencies), and how long identifiable data will be kept.
Because the Trott and Bergen (2023) dataset is publicly available under an open license, the confidentiality obligations in this study attach to the working merged dataset — the combination of the published response time data with LCSH structural variables derived by the principal investigator — rather than to the source data itself. The working merged dataset will be stored locally on a password-protected personal computer accessible only to the principal investigator. A backup copy will be maintained in a password-protected folder in the principal investigator's institutional Microsoft OneDrive account, access to which is governed by the University of Denver's data security policies. No copies of the dataset will be stored on unencrypted external drives, shared cloud folders, or any storage accessible to unauthorized parties. The faculty sponsor (Dr. Krystyna Matusiak) may review analytical outputs — summary statistics, model results, and derived variables — in her capacity as capstone advisor. She will not be provided with a copy of the full merged dataset. No sponsors, government agencies, or other external parties will have access to the data. The published source dataset contains limited indirect identifiers (participant age, self-reported gender, and native language status). These variables will be retained in the working dataset only as statistical covariates and will not be used to attempt to identify any individual. Because the source data are already publicly available and the indirect identifiers cannot, in combination, reasonably enable identification of any participant, the residual confidentiality risk associated with these variables is minimal. The working merged dataset will be retained for a minimum of three years following completion of the capstone project, consistent with standard research data retention practice, after which it may be destroyed or, if the study proceeds to publication, retained and deposited in a public repository (e.g., OSF) consistent with open science norms. Any such public deposit will contain only the variables already present in the original published dataset plus the LCSH-derived structural variables; no new identifiable information will be introduced.

3.7. Do you anticipate using these data for other studies in the future?

☒ Yes
☐ No

If answered 'YES' please explain.

If the study proceeds to publication, the anonymous working dataset — comprising the publicly available Trott and Bergen (2023) response time data merged with LCSH-derived structural variables computed by the principal investigator — may be deposited in a public repository (e.g., OSF) consistent with open science norms. No new identifiable information would be introduced in any such deposit. This possibility is disclosed to participants in the consent form's Future Research and Data Sharing section.